

Multispectral Imagery for Wind Turbine (Windmill) Applications

Project Scope Document — with Open-Source Technology Stack and Sample Outputs

Version 1.0 | 23 July 2026 | Status: Draft

1. Purpose & Objectives

This document defines the scope of a system that uses multispectral imagery (visible, red-edge, near-infrared and optionally SWIR/thermal bands) across the lifecycle of wind turbines: site selection, blade and structural inspection, environmental monitoring, and operations & maintenance (O&M) optimization. The solution is built entirely on open-source software.

- Automate detection of blade defects (erosion, cracks, moisture ingress, lightning damage) from UAV multispectral captures.
- Produce site-analysis products (NDVI, land cover, soil/water masks) from open satellite data (Sentinel-2, Landsat).
- Track vegetation and environmental change around wind farms for permit compliance.
- Deliver georeferenced defect maps, severity classifications and inspection reports through an open-source GIS/web stack.

2. Multispectral Imagery in Wind Applications

Multispectral sensors capture reflectance in discrete bands beyond human vision — typically Blue, Green, Red, Red-Edge and Near-Infrared (400–1000 nm), optionally extended with SWIR and thermal-IR. Because materials such as composite gel coat, moisture, ice, corrosion, vegetation and bare soil each have distinct spectral signatures, band ratios and indices reveal conditions that ordinary RGB imagery misses. The main application areas across the wind energy lifecycle are:

Application Area	What multispectral imagery provides
Blade inspection & damage detection	Cracks, leading-edge erosion, delamination, pitting and coating degradation; NIR/SWIR reveals moisture ingress and bonding defects under the gel coat; lightning-strike burn signatures; ice detection on blades for de-icing decisions.
Thermal & structural health monitoring	Combined with thermal-IR: hotspots in nacelle, gearbox, generator and electrical junctions; internal blade flaws via thermal gradients; corrosion on tower and flanges.
Site selection & pre-construction survey	Land cover / land use classification from Sentinel-2 or Landsat; soil moisture and terrain indicators for foundations and access roads; NDVI/EVI ecological baselines; surface roughness input to wind-flow modelling.
Environmental & ecological monitoring	Vegetation-health change detection around the farm; bird/bat habitat mapping for curtailment planning; erosion and drainage monitoring; periodic compliance reports.
Operations & maintenance optimization	Repeatable UAV fleet-wide blade scans with minimal downtime; spectral trend analysis for predictive maintenance; digital-twin integration; rapid post-storm triage.
Wake & vegetation studies	Vegetation stress patterns downstream of turbines (microclimate/wake effects); agricultural coexistence studies on leased land.
Offshore-specific applications	Marine growth / biofouling on foundations; coating breakdown in the splash zone; sediment-plume monitoring during construction (phase 2).

3. In-Scope Work Packages

WP	Work Package	Description
WP1	Data acquisition	UAV multispectral flights (blade & tower); ingest of Sentinel-2/Landsat scenes for site & environment products.
WP2	Preprocessing	Radiometric calibration, band alignment/registration, orthorectification, mosaicking, reflectance conversion.
WP3	Index & analytics engine	NDVI/NDRE/EVI computation, anomaly masks, change detection between inspection epochs.
WP4	Defect detection (ML)	CNN / classical CV models for crack, erosion, moisture and lightning-damage classification with severity grading.
WP5	GIS & reporting	Georeferenced defect layers, web map dashboard, automated PDF inspection reports.
WP6	O&M integration	Export to CMMS/digital-twin via open APIs; trend analysis across inspection cycles.

Out of scope (phase 1): SCADA data fusion, offshore marine-growth monitoring, real-time onboard (edge) inference, LiDAR integration.

4. Open-Source Software Stack

Layer	Tool (License)	Role in this project
Satellite data access	ESA Sentinel-2 / USGS Landsat (free & open data); sentinelsat, pystac-client (GPL/Apache)	Free multispectral scenes and API download for site & environmental products.
Photogrammetry / stitching	OpenDroneMap / WebODM (AGPL-3.0)	Orthomosaics and 3D reconstruction from UAV multispectral flights, incl. band alignment for MicaSense-class sensors.
Raster / geospatial core	GDAL (MIT), Rasterio (BSD), rioxarray (Apache-2.0)	Reading, reprojecting, calibrating and writing multispectral rasters.
Scientific computing	NumPy, SciPy, xarray (BSD)	Band math: NDVI, NDRE, EVI, custom anomaly indices.
Image processing / CV	OpenCV (Apache-2.0), scikit-image (BSD)	Denoising, thresholding, morphology, contour extraction for defect masks.
Machine learning	PyTorch (BSD) or TensorFlow (Apache-2.0); scikit-learn (BSD); Ultralytics YOLO (AGPL-3.0)	Defect detection/classification models; random-forest land-cover classification.
Semantic segmentation	segmentation-models-pytorch (MIT)	U-Net/DeepLab blade-surface defect segmentation.
GIS desktop	QGIS (GPL-2.0) + Semi-Automatic Classification Plugin	Visual QA, manual labelling, land-cover classification workflows.
Spatial database	PostGIS on PostgreSQL (GPL/PostgreSQL licenses)	Storage of defect geometries, turbine assets, inspection epochs.
Web mapping / dashboard	GeoServer (GPL-2.0), Leaflet (BSD), Streamlit (Apache-2.0)	Serving tiles and interactive inspection dashboards.
Visualization / reports	Matplotlib (PSF-based), ReportLab (BSD)	Figures and automated PDF inspection reports (this document was produced with them).
Workflow orchestration	Apache Airflow (Apache-2.0)	Scheduled ingestion, processing and report pipelines.

5. Sample Outputs (Synthetic Demonstration Images)

The images below are synthetic samples generated with NumPy + Matplotlib to illustrate the expected products. Production outputs will use real UAV and Sentinel-2 data processed through the stack in Section 4.

5.1 Site analysis product — multispectral bands and NDVI

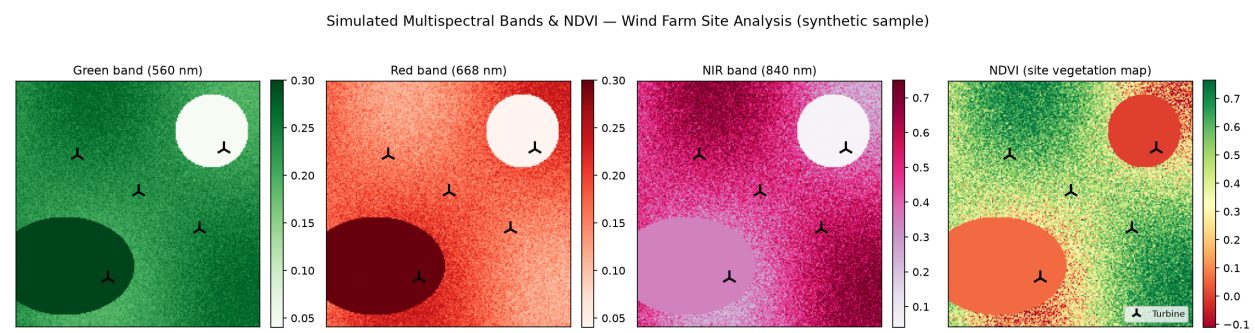


Figure 1 — Simulated green/red/NIR reflectance and derived NDVI over a wind farm site. Turbine symbols mark asset locations; low-NDVI areas (water body, bare soil) are automatically excluded from vegetation-compliance monitoring.

5.2 Blade inspection product — NIR capture and automated defect mask

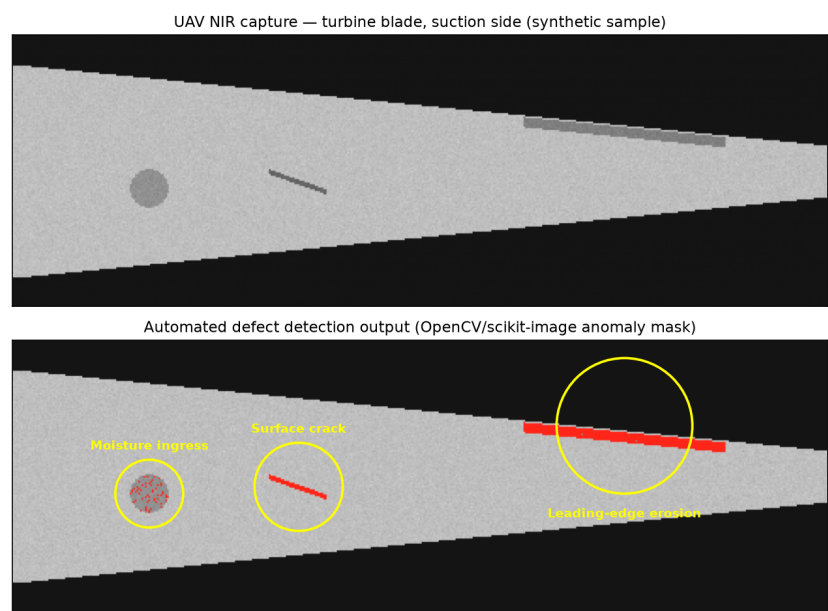


Figure 2 — Simulated UAV NIR image of a blade (top) and the automated anomaly detection overlay (bottom): leading-edge erosion, a surface crack and a moisture-ingress zone are flagged in red with yellow region-of-interest markers.

6. Processing Pipeline

Step	Stage	Open-source tools
1	Acquire — UAV flight plans / satellite scene download	OpenDroneMap ground control, sentinelsat/pystac-client
2	Calibrate — radiometric correction, reflectance panels	OpenDroneMap, Rasterio, NumPy
3	Register & mosaic — band alignment, orthomosaic	OpenDroneMap/WebODM, GDAL
4	Analyze — indices (NDVI/NDRE), anomaly masks	NumPy, xarray, OpenCV, scikit-image
5	Detect — ML defect classification & severity grading	PyTorch, segmentation-models-pytorch, YOLO
6	Store & serve — geodata layers, tiles, dashboard	PostGIS, GeoServer, Leaflet, Streamlit

Step	Stage	Open-source tools
7	Report — automated inspection PDFs, trend charts	Matplotlib, ReportLab, Airflow (scheduling)

7. Deliverables

- D1 — Calibrated multispectral orthomosaics per turbine/site (GeoTIFF).
- D2 — Defect detection layer with severity classes (GeoPackage/PostGIS).
- D3 — NDVI / land-cover / change-detection maps for environmental compliance.
- D4 — Web dashboard (Leaflet/Streamlit) with per-turbine inspection history.
- D5 — Automated PDF inspection reports (per flight campaign).
- D6 — Reproducible open-source pipeline (containerized, Apache Airflow DAGs).

8. Standards, Assumptions & Constraints

- Inspection practice aligned with IEC 61400 series and DNV-GL blade inspection guidance.
- UAV operations subject to local aviation (e.g. DGCA/FAA/EASA) rules; flights only in permitted wind conditions.
- Multispectral sensor with at least 5 bands (B, G, R, RedEdge, NIR); SWIR/thermal optional add-ons.
- All software components are OSI-approved open source; AGPL components (WebODM, YOLO) used as services to keep licensing obligations contained.
- Sample images in this document are synthetic and for illustration only.